Evidence-Linked Clinical Trial Criterion Screening with a Locally Executable Transformer

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Individual student research project

Abstract—This work implements an end-to-end research prototype that predicts TrialGPT criterion-level eligibility labels and cites source sentences from patient descriptions. We fine-tuned a pinned Qwen2.5-1.5B-Instruct checkpoint using LoRA on the prescribed patient-grouped TrialGPT Criterion Annotations training split. The held-out split contains 206 annotations from eight patients and was evaluated once after a checksummed experiment freeze. Relative to the same base checkpoint, the adapted model improved test accuracy from 0.146 to 0.704 and macro-F1 from 0.137 to 0.362. Evidence micro-F1 increased from 0.165 to 0.296. The adapted model nevertheless remains weak on minority labels, and its output should not be interpreted as a clinical enrollment decision. We also implement an explicit key/value-cache decoder, tiled online-softmax attention, and greedy speculative decoding with a 0.5B draft model. Manual cache decoding matched the uncached reference in all six adapter-backed cases; speculative output matched in 36/36 cases, but local speculation was slower than cached target decoding. A local browser application exposes criterion results, exact cited text, review flags, exports, and a live attention comparison.

Index Terms—clinical trial matching, evidence extraction, LoRA, key/value cache, online softmax, speculative decoding

I. Introduction

Screening a patient record against a trial criterion requires more than topical similarity: inclusion and exclusion polarity, missing information, and evidence location all matter. TrialGPT motivates criterion-level decomposition and evidence inspection, but its published results are not comparable to this assignment's patient-grouped split [1]. We implement an independently trained, locally executable prototype using the required TrialGPT Criterion Annotations dataset. Our objective is to expose both useful predictions and the operational limits visible in held-out results.

The contribution is a reproducible pipeline with (1) four-way criterion labels and original sentence IDs; (2) a validation-selected LoRA adapter evaluated against the same untouched test rows as its base model; (3) an examined cache-aware greedy loop, online-softmax attention implementation, and target-verified speculative decoder; and (4) a standalone local browser application. This is a criterion-level research result. The aggregate patient-level status is a transparent rule-based aid, not a labeled or validated trial outcome.

II. Related Work

TrialGPT decomposes eligibility matching into interpretable criterion-level operations and motivates the source-linked output used here [1]. Qwen2 technical documentation describes the decoder family; we pin exact target and draft checkpoint revisions rather than infer properties from a model-family name [2]. LoRA freezes pretrained weights and learns low-rank updates [3]; it reduces trainable parameters but does not eliminate inference activations or guarantee task improvement. FlashAttention and online attention work show how tiling and stable rescaling can reduce intermediate memory traffic, while a Python implementation should not be expected to match a fused kernel [4], [5]. Speculative decoding verifies draft proposals with a target model, preserving target output under appropriate acceptance and cache rules [6]. We implement deterministic greedy verification and measure its local cost rather than assuming a published speedup transfers to our hardware.

III. Materials and Methods

A. Dataset and protocol

We use the required TrialGPT Criterion Annotations release pinned to revision 1cfcafde94a1560a33b4addc1638664fe28fc059. Patient IDs are sorted and shuffled with NumPy default_rng(42); the prescribed 37/8/8 patient split yields 602 prepared training rows, 206 validation rows, and 206 test rows. One known missing-criterion training row (annotation 883) is quarantined under the documented decision. Data audits confirm patient disjointness and exclude model-generated predictions and explanations from inputs. The final test labels were accessed only by the guarded evaluator after the frozen manifest was written.

Each example contains the patient sentences, criterion type, and criterion text. The prompt asks for JSON with a label from the preserved inclusion or exclusion label set and integer source-sentence IDs. Whole leading sentences are retained under the 1024-token input budget; the output cap is 96 tokens. The parser validates allowed labels and IDs; the application resolves citations to exact source text. No evidence is not treated as proof of ineligibility, and a transparent aggregation forces review for uncertain, invalid, contradictory, or truncated outputs.

B. Models and adaptation

The target is Qwen/Qwen2.5-1.5B-Instruct, revision 989aa7980e4cf806f80c7fef2b1adb7bc71aa306, with the Qwen2 causal decoder and pinned tokenizer. The

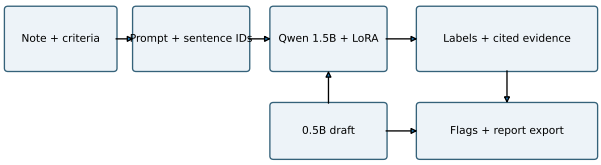


Fig. 1. End-to-end data flow. The final application loads one adapted target; speculative mode additionally loads the pinned draft. Exact cited sentences and review flags remain visible.

smaller speculative draft is Qwen/Qwen2.5-0.5B-Instruct, revision 7ae557604adf67be50417f59c2c2f167def9a775; vocabulary equality was checked. Both checkpoints use Apache-2.0 model licenses. LoRA targets query and value projections with rank 8, alpha 16, and dropout 0.05. Training uses AdamW at 2×10^{-4}, microbatch 1, gradient accumulation 8, gradient checkpointing, FP16, and three epochs. The full run completed 228 optimizer steps on a Colab Tesla T4. Epoch 3 was selected using validation macro-F1 with evidence F1 and validation loss tie-breaks. The adapter was verified locally on an RTX 3050 Laptop GPU with 6 GiB.

C. Decoding and attention

The canonical generation contract is deterministic greedy decoding with one beam, pinned EOS/pad IDs, and the target generation-config repetition penalty. The uncached reference calls generate(use_cache=False) so processors remain active. The explicit decoder prefills once, then forwards one committed token at a time with absolute cache positions, a growing attention mask, and cache-length assertions. The speculative path drafts up to k tokens, verifies the block with target logits, accepts the matching prefix, appends the target correction token at the first mismatch, and crops both caches to the committed length. Qwen's history-dependent repetition penalty is applied to both draft and target token histories.

For attention, the dense reference computes scaled dot products in FP32 and handles boolean visibility masks and fully masked rows. The tiled implementation processes query and key blocks while maintaining the online maximum, normalizer, and weighted-value accumulator. It avoids materializing the full score matrix, but its Python loops incur overhead. For a score tile with row maximum m_t, normalizer ℓ_t, and weighted sum o_t, a new tile with maximum m is merged using $m\prime = \max(m_t, m)$, $\ell\prime = e^{m_t - m\prime}\ell_t + \sum_j e^{s_j - m\prime}$, and $o\prime = e^{m_t - m\prime}o_t + \sum_j e^{s_j - m\prime}v_j$; the final output is o/ℓ.

IV. Experimental Setup and Metrics

Classification uses accuracy, fixed-class macro-F1, per-class precision/recall/F1, and confusion matrices, overall and by criterion type. Evidence uses micro precision, recall, and F1 over sentence-ID sets; empty gold evidence contributes no true positives and predicted IDs remain false positives. We report overall and inclusion/exclusion results. Tests use the same frozen IDs, prompt, parsing, metric, and generation contract for base and adapted variants.

Local measurements use an RTX 3050 Laptop GPU, CUDA-enabled PyTorch 2.11.0, Transformers 4.57.1, and FP16. Training ran separately on a Colab Tesla T4. Decoder correctness compares token IDs exactly. Timings are local and should not be compared across these hardware environments. The attention experiment uses synthetic tensors and does not claim a transformer-kernel speedup.

TABLE I
Final test results on the frozen 206-row split

Task	Metric	Base	Adapted
Classification	Accuracy	0.146	0.704
	Macro-F1	0.137	0.362
Evidence	Micro precision	0.091	0.186
	Micro recall	0.926	0.723
	Micro-F1	0.165	0.296

V. Results

A. Held-out criterion performance

Table I reports the paired final test results. Fine-tuning raises accuracy and macro-F1 while leaving substantial error, particularly among minority labels. The model collapses much of the exclusion task to "not excluded"; this explains why accuracy alone would give an incomplete view. Evidence recall is higher than evidence precision, reflecting overinclusive citations. The app displays exact cited text and never presents an aggregate preliminary match as ground truth.

B. Per-type results and errors

Inclusion and exclusion results are separated in Table II. We provide confusion matrices and a case-level ledger in the generated evidence package. Eight representative errors use eight distinct test patients. Examples include an incorrect "not enough information" response to explicit shock-related evidence and an unsupported "not excluded" response for criteria with no supporting source. The error ledger preserves the gold/predicted evidence excerpts and raw structured output. These examples demonstrate category confusion and evidence overreach; they do not establish why the model produced an answer.

C. Inference mechanisms

The adapter-backed explicit cache path matches the uncached reference in 6/6 validation cases over three prompt lengths and two output caps. The adapter-backed speculative target matched exactly in 36/36 validation cases over three prompts, caps 8/16/32, and $k = 1/2/4/8$. On one 356-token validation prompt with a 16-token cap, ordinary cached target generation took 0.930 s median.

TABLE II
Test metrics by criterion type

Type	Metric	Base	Adapted
Inclusion	Accuracy	0.253	0.720
	Macro-F1	0.191	0.382
	Evidence micro-F1	0.286	0.485
Exclusion	Accuracy	0.084	0.695
	Macro-F1	0.050	0.205
	Evidence micro-F1	0.081	0.142

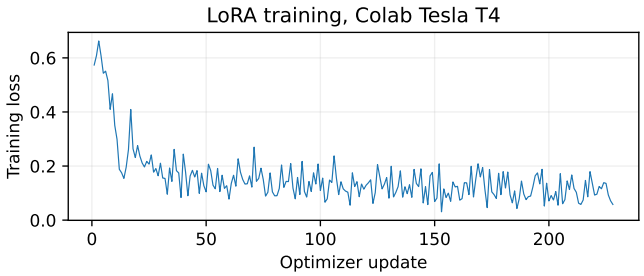


Fig. 2. Training loss across 228 optimizer updates. Validation selected epoch 3; this curve is not a test-set tuning signal.

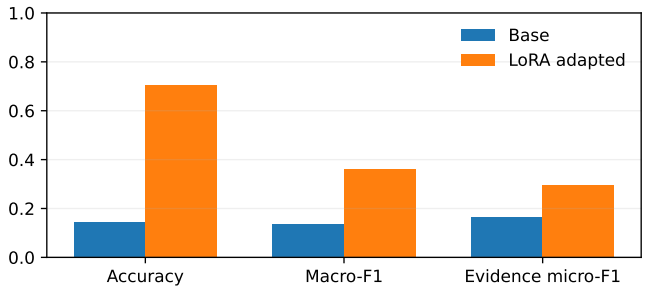


Fig. 3. Paired held-out accuracy, macro-F1, and evidence micro-F1. Fine-tuning improves all three aggregate measures on the frozen split.

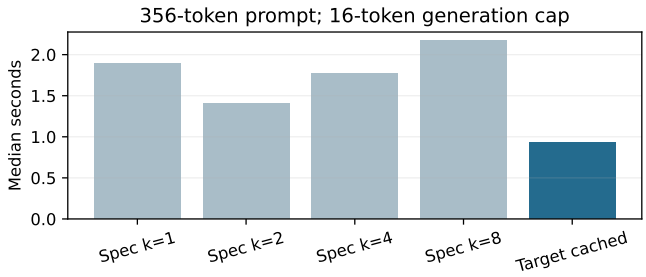


Fig. 4. Adapter-backed speculative decoding latency versus cached target greedy decoding on one validation prompt. Lower is faster; target-only is the reference.

Speculative $k = 2$ took 1.402 s (0.664× speedup, i.e., slower); $k = 8$ took 2.167 s (0.429×). The 0.5B draft was resident alongside the 1.5B target, with peak allocated memory around 4.22–4.34 GB. Correctness is established for this sample; a local speedup is not.

For the base target, explicit KV reuse was 2.6–3.0× faster than the uncached reference on the saved local prompts. With the selected adapter, explicit caching was 2.19× and 2.94× faster on 335- and 356-token validation prompts, respectively, at a 16-token cap (five repetitions per mode). The attention benchmark compared dense and tiled methods across sequence lengths 128–2048 and tile widths 16/32/64, with five synchronized repetitions. Numerical errors remained small, but the simple tiled implementation was slower because Python launches and loop overhead dominate. Figure 2 summarizes adapter training loss; Figure 3 compares the final test metrics.

VI. Limitations and Discussion

The benchmark has 206 criterion annotations from only eight test patients. Label supports are imbalanced, and the adapted model has zero test support for some classes it fails to predict. The results are not clinical validation, external replication, or a trial-level eligibility assessment. Evidence matching is sentence-level set overlap; it does not prove entailment, temporal interpretation, or causal reasoning. The local speculative benchmark uses one prompt and five repetitions, so it supports a hardware-specific slowdown observation rather than a general efficiency claim. The tiled attention kernel is educational Python code, not fused production attention. Colab and laptop results are reported separately. The app is local and offline-capable once model weights and dependencies are present; speculative mode needs additional memory.

VII. Conclusion

We completed LoRA adaptation and an adapter-backed local screening application, then evaluated base and adapted models once on the same frozen test annotations. Fine-tuning substantially improved aggregate criterion metrics, while severe minority-class and evidence-localization errors remain. The cache-aware decoder, online-softmax attention, and greedy speculative implementation are inspectable and supported by correctness and timing artifacts. The application exposes source-linked predictions, uncertainty, runtime modes, and downloadable reports with a clear educational notice. These results support a demonstration of the implemented system, not clinical use.

AI-Use Appendix

AI assistants were used to help plan implementation, draft and revise code/documentation, and review work. The author ran the experiments, inspected the repository and outputs, and is responsible for verifying, explaining, and defending the submitted work. No assistant-generated prediction or explanation was included as a model input feature.

TABLE III
Per-class test precision, recall, F1, and support.

Type	Variant	Class	Precision	Recall	F1	Support
Inclusion	Base	included	0.73	0.59	0.65	27
Inclusion	Base	not applicable	0.06	1.00	0.11	3
Inclusion	Base	not enough information	0.00	0.00	0.00	38
Inclusion	Base	not included	0.00	0.00	0.00	7
Inclusion	Adapted	included	1.00	0.59	0.74	27
Inclusion	Adapted	not applicable	0.00	0.00	0.00	3
Inclusion	Adapted	not enough information	0.64	1.00	0.78	38
Inclusion	Adapted	not included	0.00	0.00	0.00	7
Exclusion	Base	excluded	0.00	0.00	0.00	0
Exclusion	Base	not applicable	0.11	0.92	0.20	12
Exclusion	Base	not enough information	0.00	0.00	0.00	28
Exclusion	Base	not excluded	0.00	0.00	0.00	91
Exclusion	Adapted	excluded	0.00	0.00	0.00	0
Exclusion	Adapted	not applicable	0.00	0.00	0.00	12
Exclusion	Adapted	not enough information	0.00	0.00	0.00	28
Exclusion	Adapted	not excluded	0.69	1.00	0.82	91

TABLE IV
Test confusion matrices by criterion type and model variant. Rows are gold; columns are predicted.

Inclusion / Base Gold / Pred.	included	not included	not enough information	not applicable	___invalid___	Support
included	16	0	0	11	0	27
not included	3	0	0	4	0	7
not enough information	3	0	0	34	1	38
not applicable	0	0	0	3	0	3
___invalid___	0	0	0	0	0	0
Inclusion / Adapted **Gold / Pred.**	**included**	**not included**	**not enough information**	**not applicable**	**___invalid___**	**Support**
included	16	0	11	0	0	27
not included	0	0	7	0	0	7
not enough information	0	0	38	0	0	38
not applicable	0	0	3	0	0	3
___invalid___	0	0	0	0	0	0
Exclusion / Base **Gold / Pred.**	**excluded**	**not excluded**	**not enough information**	**not applicable**	**___invalid___**	**Support**
excluded	0	0	0	0	0	0
not excluded	27	0	0	64	0	91
not enough information	6	0	0	22	0	28
not applicable	1	0	0	11	0	12
___invalid___	0	0	0	0	0	0
Exclusion / Adapted **Gold / Pred.**	**excluded**	**not excluded**	**not enough information**	**not applicable**	**___invalid___**	**Support**
excluded	0	0	0	0	0	0
not excluded	0	91	0	0	0	91
not enough information	0	28	0	0	0	28
not applicable	0	12	0	0	0	12
___invalid___	0	0	0	0	0	0

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TABLE V
Eight representative label errors from eight distinct held-out patients. Evidence excerpts are exact source spans, shortened only at display boundary.

Annotation / type	Criterion	Gold to predicted	Gold evidence excerpt	Predicted evidence excerpt
161 / Inclusion	Patients undergoing elective carotid endarterectomy	not included → not enough information	(no cited sentence IDs)	A physician is called to see a 67-year-old woman...
185 / Inclusion	Patients were included if they were diagnosed with Shock, ...	included → not enough information	On examination she is pale and in moderate disco...	A 40-year-old woman with no past medical history...
315 / Exclusion	BMI> 40 kg/m2.	not enough information → not excluded	(no cited sentence IDs)	A 48-year-old white male with history of common ...
382 / Inclusion	Fulfil at least 4 criteria for SLE as defined by the Ameri...	included → not enough information	A 21-year-old female is evaluated for progressiv...	A 21-year-old female is evaluated for progressiv...
418 / Exclusion	Failure to pass MRI metal screening	not enough information → not excluded	(no cited sentence IDs)	On an abdominal ultrasound, a markedly edematous...
682 / Inclusion	Primary indication for AT is nocturnal obstructive symptom...	included → not enough information	A 10 yo boy with nighttime snoring, pauses in br...	A 10 yo boy with nighttime snoring, pauses in br...
745 / Inclusion	female patients have adequate birth-control method	not applicable → not enough information	(no cited sentence IDs)	A 5-year-old boy presents to the emergency depar...
834 / Inclusion	–Negative pregnancy test for females	not applicable → not enough information	(no cited sentence IDs)	A 4 year old boy presents to the emergency room ...

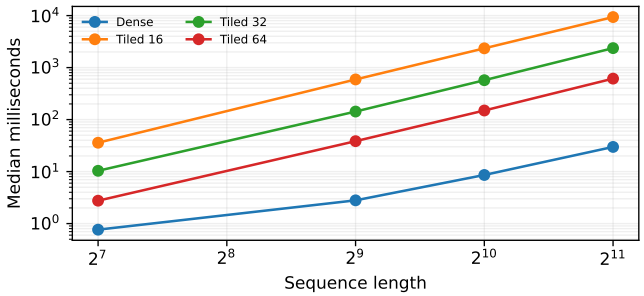


Fig. 5. Dense and tiled online-softmax attention median latency on the RTX 3050 Laptop GPU. Five timed repetitions per setting; values are not transformer inference timings.

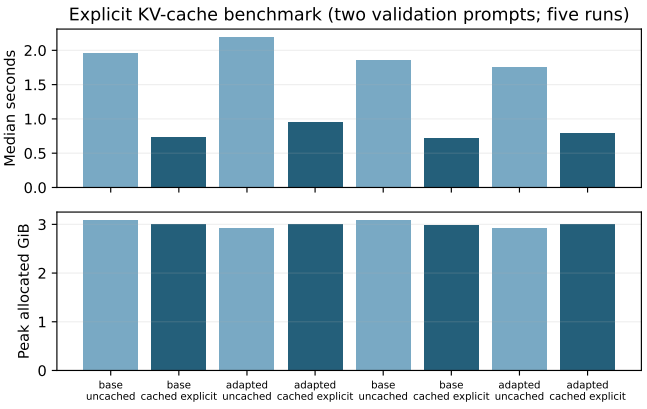


Fig. 6. Adapted cache path versus uncached generation on two validation prompts. Bars show median latency and peak allocated memory over five repetitions on the local RTX 3050.

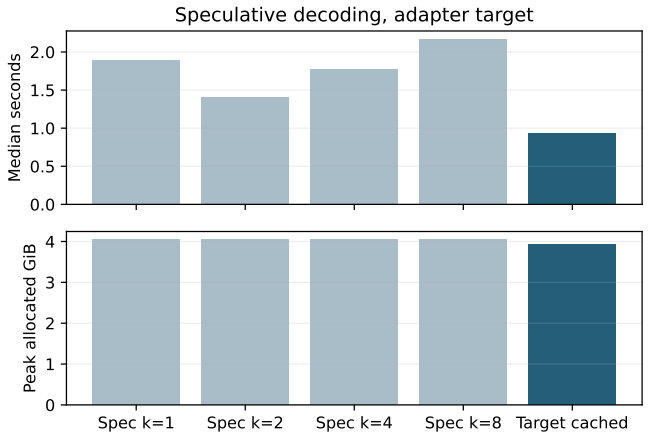


Fig. 7. Adapter-backed speculative latency and peak memory with the target and draft resident. Five repetitions on one validation prompt; correctness was exact for every measured mode.

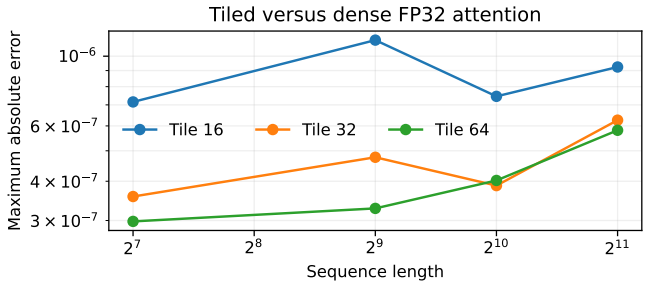


Fig. 8. Maximum absolute output error for tiled online-softmax attention against the dense reference across sequence lengths and tile widths.

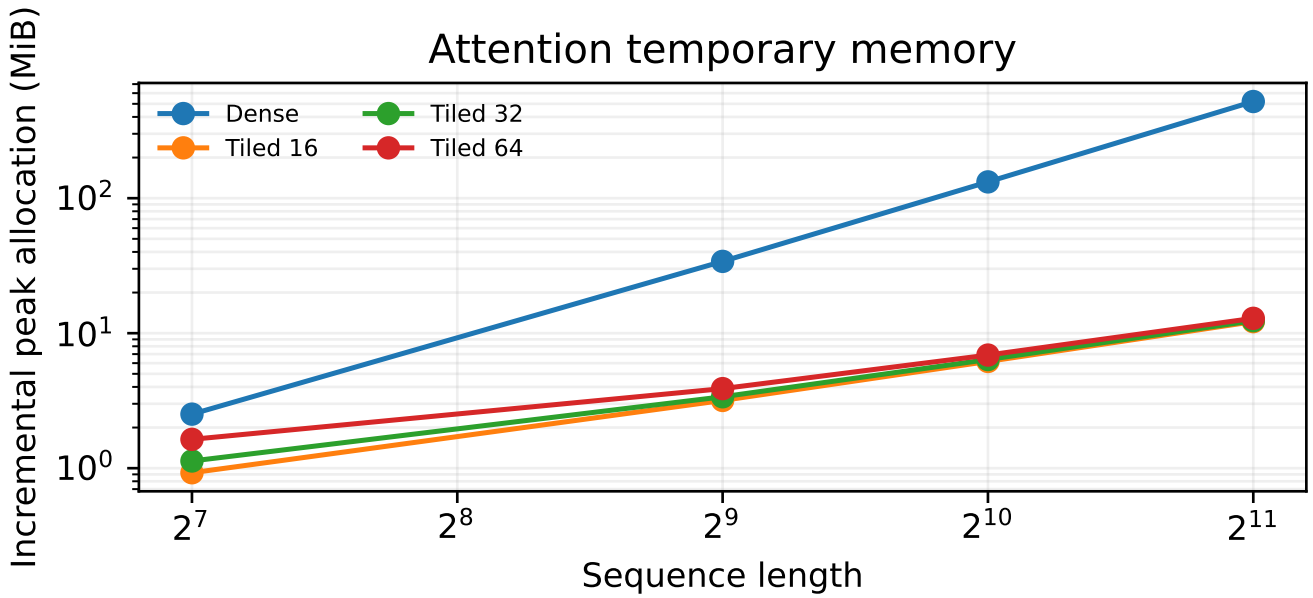


Fig. 9. Incremental peak allocation for the dense and tiled attention routines. At length 2,048, tile width 64 reduced measured incremental peak allocation from approximately 520 MiB to 13 MiB.